Physics 2110: Midterm Exam II

(Dated: October 23, 2009)

Each problem, including the extra-credit one, is worth 10 points. The extra-credit problem is due in class on Monday, Oct. 26, at 9:40 am.

1. The system shown in Fig.1 can be used to measure the acceleration of the system. An observer riding on a platform measures the angle θ that the tread supporting the light ball makes with the vertical. (a) How is θ related to the acceleration of the system if there no friction anywhere? (b) If $m_1 = 250$ kg and $m_2 = 1250$ kg, what is θ? (c) If you can vary m_1 and m_2, what is the largest angle θ you could achieve? (d) How the result in parts (a) and (b) changes if you account for the frictional force (coefficient of kinetic friction $\mu_K = 0.200$) between the platform and the horizontal surface?

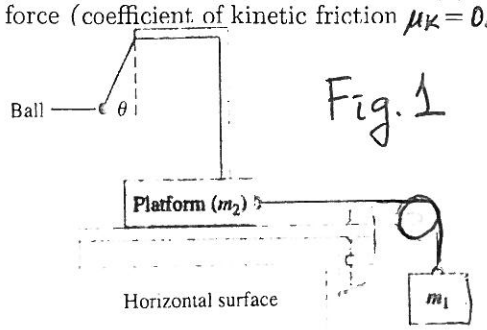


Fig. 1

2. A box is sliding with a speed $v_0 = 4.50$ m/s on a horizontal surface, when, at point P, it encounters a rough section. (a) Use the work-kinetic energy theorem to find how far this box slides before stopping if the coefficient of friction on the rough section is $\mu_k = 0.100$. (b) Imagine now that the coefficient of friction is not constant and instead increases with distance x from point P as $\mu_k(x) = 0.100 + \alpha x$, where $\alpha = 0.040\ m^{-1}$. How far the box slides before stopping in this case?

3. A small block of mass $m_1 = 0.500$ kg is released from rest at the top of a curved wedge of mass $m_2 = 3.00$ kg, which sits on a frictionless horizontal surface as in Fig.2. When the block leaves the wedge, its velocity is measured to be $v_1 = 4.00$ m/s to the right, as shown in Fig.2. (a) What is the velocity v_2 of the wedge after the block reaches the horizontal surface? (b) What is the height h of the wedge?

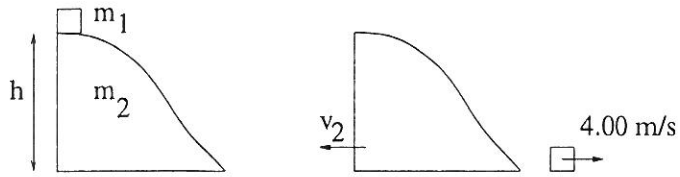
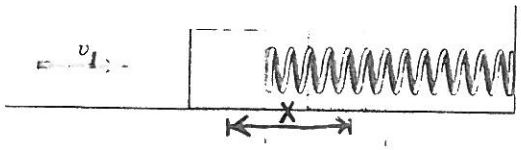


FIG. 2

4. A bullet with mass $m_1 = 0.008$ kg strikes and embeds itself in a block with mass $m_1 = 0.992$ kg that rests on a frictionless, horizontal surface and is attached to a coil spring (Figure 3) with stiffness $k = 5.00$ N/m. Find the maximum compression of the spring after the impact if the initial velocity of the bullet is $v_1 = 325$ m/s.

Fig. 3



5: extra credit problem. A very light tennis ball of mass m_1 is placed on top of a basketball of mass m_2. The balls are released from the height of $h_0 = 1.5$ m and fall freely (neglect air resistance) until they hit the ground, see Fig.4. Find the maximum height to which the tennis ball rises after bouncing off the basketball. All collisions are elastic. Assume also that $m_2 \gg m_1$ and, in addition, treat the balls as point objects.
Hint: during the downward motion of the balls consider that the basketball is well in front of the tennis ball and hits the ground first.

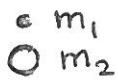


Fig. 4